Ch 10.1: Neural Nets

Lecture 28 - CMSE 381

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Announcements

Last time:

- SVM

This lecture:

- Feed Forward Neural Nets

Announcements:

- HW # 9 tonight

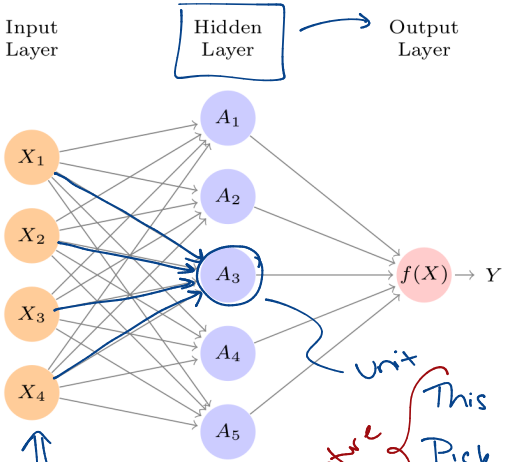
Section 1

Neural Nets

The idea

$$\begin{matrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{matrix} \rightsquigarrow \boxed{f(X)} \rightsquigarrow Y$$

Feed Forward Neural Network: The cartoon



$p = 4$

Input data

unit

Each arrow represents taking a coef x the number

All arrows in together represent taking a linear combination

$$A_3 = g(c_0 + c_1 X_1 + c_2 X_2 + c_3 X_3 + c_4 X_4)$$

architecture

This example: One hidden layer

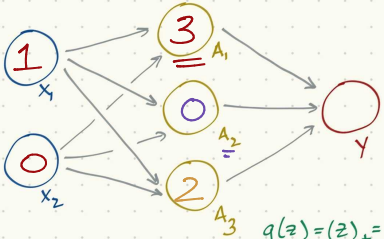
Pick how many hidden units are in the layer $K = 5$

A very simple example

Computing A_k for $(1, 0)$

g is the "activation function"

$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$



$$g(z) = (z)_+ = \begin{cases} z & \text{if } z \geq 0 \\ 0 & \text{else} \end{cases}$$

w matrix

	0	1	2
1	1	2	1
2	-1	0	1
3	3	-1	-1

β matrix

$$\begin{array}{c} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{array} \begin{pmatrix} -1 \\ 2 \\ 1 \\ -2 \end{pmatrix}$$

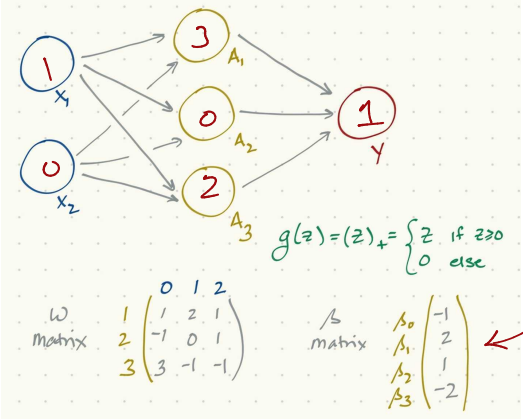
$$A_1 = g(w_{10} + w_{11} X_1 + w_{12} X_2)$$
$$= g(1 + 2 \cdot (1) + 1 \cdot (0))$$
$$= g(3)$$
$$= 3$$

$$A_2 = g(-1 + 0(1) + 1(0))$$
$$= g(-1) = 0$$

A very simple example

Computing Y for $(1, 0)$

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$



$$\beta_0 + \beta_1 A_1 + \beta_2 A_2 + \beta_3 A_3$$

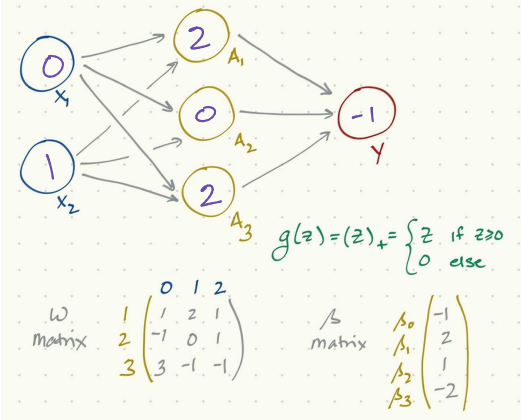
$$-1 + 2(3) + 1(0) + (-2)(2)$$

$$-1 + 6 + 0 - 4 = 1$$

A very simple example

Computing Y for $(0, 1)$

$$A_k = h_k(X) = g(w_{k0} + \textstyle\sum_{j=1}^{p} w_{kj} X_j), \qquad f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$



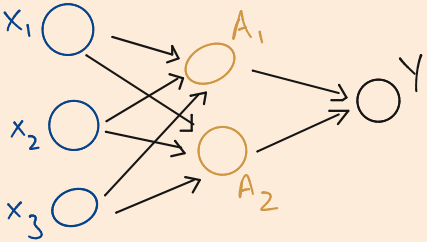
$$-1 + 2(2) + 1(0) + (-2)(2)$$

$$-1 + 4 - 4 = -1$$

A different example

- Draw the diagram for a neural net with input data points with $p = 3$ (i.e., (X_1, X_2, X_3)) and two units in the hidden layer.
- Using the ω and β matrices, what is the output predicted Y for the point $(2, 0, 1)$?

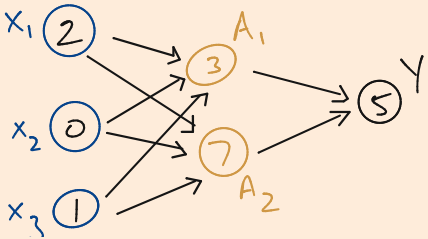
$$\omega = \begin{pmatrix} 1 & 0 & -2 & 2 \\ 3 & 1 & 0 & 1 \end{pmatrix} \beta = \begin{pmatrix} -1 \\ 2 \\ 0 \\ \cancel{1} \end{pmatrix}$$



- Use the activation function

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{else.} \end{cases}$$

Extra space



$$\omega = \begin{pmatrix} 1 & 0 & -2 & 2 \\ 3 & 1 & 0 & 1 \end{pmatrix} \quad \beta = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}$$

$$A_1 = 1 + 0(2) + (-2)(0) + 2(1)$$
$$= 3$$

$$A_2 = 3 + 1(2) + 0(0) + 1(1)$$
$$= 6$$

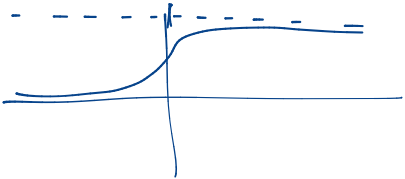
$$Y = -1 + 2(3) + 0(6)$$
$$= -1 + 6 = 5$$

Choices for activation function

Sigmoid:

$$g(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

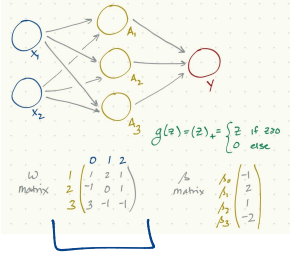
Converts linear input to probabilities in [0,1]



ReLU: Rectified linear unit

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{else.} \end{cases}$$

Matrix version: First layer

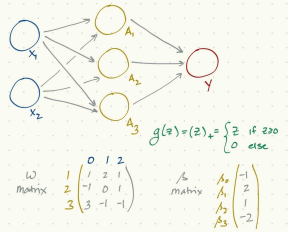


$$A_k = h_k(X) = g(w_{k0} + \textstyle\sum_{j=1}^{p} w_{kj} X_j),$$

$$A = g(\mathbf{W} \cdot \mathbf{X}) \qquad \mathbf{X}^T = (1\ X_1\ X_2\ \cdots\ X_p)$$

$$\underbrace{\begin{pmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \\ 3 & -1 & -1 \end{pmatrix}}_{W} \underbrace{\begin{pmatrix} 1 \\ X_1 \\ X_2 \end{pmatrix}}_{X} = \underbrace{\begin{pmatrix} 1 + 2X_1 + 1 \cdot X_2 \\ -1 + 0X_1 + 1X_2 \\ 3 + (-1)X_1 + (-1)X_2 \end{pmatrix}}$$

Matrix version: Output



$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

$$Y = \beta \cdot \mathbf{A}^{(\cancel{2})} \qquad (\mathbf{A}^{(\cancel{2})})^T = (1 \; A_1^{(\cancel{2})} \; A_2^{(\cancel{2})} \; \cdots \; A_K^{(\cancel{2})})$$

$$\begin{pmatrix} -1 & 2 & 1 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ A_1 \\ A_2 \\ A_3 \end{pmatrix} = -1 + 2A_1 + 1A_2 + -2A_3$$

Now what?

Choose parameters by minimizing RSS, $\sum_{i=1}^{n}(y_i - f(x_i))^2$

Chosen in advance:

layers

hidden units

Activation function

Tuned by the model:

$W = (w_{ij})$

$\beta = (\beta_0, \ldots, \beta_K)$

Coding

Next time

20	F	Nov 4	Polynomial & Step Functions.	7.1,7.2	
21	M	Nov 7	Step Functions	7.2	
22	W	Nov 9	Basis functions, Regression Splines	7.3,7.4	
23	F	Nov 11	Decision Trees	8.1	HW #7 Due
24	M	Nov 14	Random Forests	8.2.1, 8.2.2	
25	W	Nov 16	Maximal Margin Classifier	9.1	
26	F	Nov 18	SVC	9.2	HW #8 Due
27	M	Nov 21	SVM	9.3, 9.4, 9.5	
28	W	Nov 23	Extended virtual office hours		
	F	Nov 25	No class - Thanksgiving		
29	M	Nov 28	Single layer NN	10.1	HW #9 Due
30	W	Nov 30	Multi Layer NN	10.2	
31	F	Dec 2	CNN	10.3	
32	M	Dec 5	Unsupervised Learning & Clustering	12.1, 12.4	HW #10 Due
	W	Dec 7	Review		
	F	Dec 9	Midterm #3	Bring your cheat sheet and a non-internet-connected calculator	